Mark Scheme

B卷

PHYSICS · Paper 4 (A2)

A2 Mid-term Examination — 5 questions · 100 marks

Mark scheme — Question 1

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Question	Answer	Marks
2(a)	horizontal force on sphere causes centripetal acceleration	B1
	weight of sphere is (now) equal to vertical component of tension **or** horizontal and vertical components (of force) (now) combine to give greater tension (in spring)	B1
	greater tension in spring so greater extension of spring	B1
2(b)(i)	$r = 10.8 \times \sin 27° = 4.9$ cm	A1
2(b)(ii)	$T \cos \theta = mg$ **or** $T \cos \theta = W$ **and** $W = mg$	C1
	$T \cos 27° = 0.29 \times 9.81$ leading to $T = 3.2$ N	A1
2(b)(iii)	$\Delta T = 3.2 - (0.29 \times 9.81)$	C1
	$k = \Delta T / \Delta x$ $= [3.2 - (0.29 \times 9.81)] / [10.8 - 8.5]$ $= 0.15$ N cm^{-1}	A1
2(c)(i)	centripetal acceleration $= (T \sin \theta) / m$ $= (3.2 \times \sin 27°) / 0.29$	C1
	$= 5.0$ m s^{-2}	A1

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Mark scheme — Question 1

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Question	Answer	Marks
2(c)(ii)	$a = r\omega^2$ **and** $\omega = 2\pi / T$ **or** $a = v^2 / r$ **and** $v = 2\pi r / T$	**C1**
	$T = 2\pi \times \sqrt{(0.049 / 5.0)}$ $= 0.62$ s	**A1**

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Mark scheme — Question 2

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Question	Answer	Marks
1(a)	at least 4 straight radial lines to P 	B1
	all arrows pointing along the lines towards P	B1
1(b)	**Any 2 from:** gravitational force provides the centripetal force (centripetal or gravitational) force has constant magnitude (centripetal or gravitational) force is perpendicular to velocity (of moon) / direction of motion (of moon)	B2
1(c)(i)	$\frac{GMm}{r^2} = mr\omega^2$	M1
	$M = \frac{r^3\omega^2}{G}$ and gradient = $r^3\omega^2$ hence $M = \frac{\text{gradient}}{G}$ or $r^3 = GM \times 1/\omega^2$ so gradient = GM hence $M = \frac{\text{gradient}}{G}$	A1
1(c)(ii)	$M = 4.1 \times 10^{23} / (6.0 \times 10^7 \times 6.67 \times 10^{-11}) = \mathbf{1.0 \times 10^{26}}$ kg	B1

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Mark scheme — Question 2

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Question	Answer	Marks
1(c)(iii)	$\frac{GMm}{r^2}=\frac{mv^2}{r}$ $\frac{GM}{r}=v^2$	C1
	$v^2=\frac{6.67\times10^{-11}\times1.0\times10^{26}}{1.2\times10^{8}}$ $v^2=5.6\times10^{7}\ m\ s^{-1}$	C1
	$v=7500\ m s^{-1}$	A1

Question	Answer	Marks
2(a)	0	B1
2(b)	$pV = nRT$ $(n =)\ 1.5\times10^{5}\times4.2\times10^{-3}/8.31\times540$	C1
	= **0.14** mol	A1
2(c)	missing pressure **1.5** ($\times 10^{5}$)	B1
	both missing volumes **1.8** ($\times 10^{-3}$)	B1
2(d)(i)	(ΔU:) increase in internal energy (of the system)	B1
	(q:) thermal energy supplied to the system	B1
	(W:) work done on system	B1

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Mark scheme — Question 3

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Question	Answer	Marks
1(a)	(gravitational) force is (directly) proportional to product of masses	B1
	force (between point masses) is inversely proportional to the square of their separation	B1
1(b)	$GMm / R^2 = mR\omega^2$	M1
	$\omega = 2\pi / T$ and algebra leading to $4\pi^2R^3 = GMT^2$	A1
	or	
	$GMm / R^2 = mv^2 / R$	(M1)
	$v = 2\pi R / T$ and algebra leading to $4\pi^2R^3 = GMT^2$	(A1)
1(c)	$4\pi^2 \times R^3 = 6.67 \times 10^{-11} \times 5.98 \times 10^{24} \times (24 \times 60 \times 60)^2$ ($R = 4.22 \times 10^7$ m)	C1
	$h = R - (6.37 \times 10^6)$	C1
	$h = (4.22 \times 10^7) - (6.37 \times 10^6)$ $= 3.6 \times 10^7$ m	A1
1(d)(i)	$\omega = 2\pi / T$	C1
	$= 2\pi / (24 \times 60 \times 60)$ $= 7.3 \times 10^{-5}$ rad s^{-1}	A1
1(d)(ii)	orbit is from east to west	B1
	orbit is not equatorial / orbit is polar	B1

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Mark scheme — Question 4

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Question	Answer	Marks
4(a)(i)	x_0 = 8.0 cm	A1
4(a)(ii)	$\omega = 2\pi / T$	C1
	$= 2\pi / 4.0 = 1.6$ rad s^{-1}	A1
4(a)(iii)	$E = \frac{1}{2}m\omega^2 x_0^2$	C1
	$= \frac{1}{2} \times 36 \times 1.6^2 \times 0.080^2$	C1
	= 0.29 J	A1
4(b)	dome-shaped curve, starting and ending at $E_K = 0$	B1
	maximum E_K shown as 0.29 J	B1
	position of peak shown at h = 10.0 cm	B1
	line intercepts h-axis at h = 2.0 cm and at h = 18.0 cm	B1

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Mark scheme — Question 5

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Question	Answer	Marks
4(a)(i)	amplitude $= ½ \times 7.2 \times 10^{-15}$ $= 3.6 \times 10^{-15}$ m	A1
4(a)(ii)	$\omega = 2\pi / (0.20 \times 10^{-6})$ $= 3.1 \times 10^{7}$ rad s^{-1}	A1
4(a)(iii)	$v_0 = \omega x_0$	C1
	$v_0 = 3.1 \times 10^{7} \times 3.6 \times 10^{-15} = 1.1 \times 10^{-7}$ m s^{-1}	A1
4(b)(i)	$I_0 = nAv_0e$ $= 8.5 \times 10^{28} \times 4.3 \times 10^{-4} \times 1.1 \times 10^{-7} \times 1.60 \times 10^{-19}$	C1
	$= 0.64$ A	A1
4(b)(ii)	sketch: two cycles of sinusoidal curve of amplitude I_0 and period 0.20 μs	B1
	correct phase, with $I = +I_0$ at $t = 0$	B1
4(b)(iii)	equation of form $I = I_0 \cos \omega t$	M1
	value of I_0 used matches answer to **(b)(i)** and value of ω used matches answer to **(a)(ii)** [if **(a)(ii)** and **(b)(i)** correct then $I = 0.64 \cos (3.1 \times 10^{7}\ t)$]	A1
4(b)(iv)	$I_{r.m.s.} = I_0 / \sqrt{2}$ $= 0.64 / \sqrt{2}$ $= 0.45$ A	A1

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